

Homework: By 19th September 2026

$$\frac{\partial^2 \phi}{\partial x^2} = 2x^2 - \frac{x}{2} + \exp(x) \quad \left. \begin{array}{l} \text{Pen \& Paper} \\ \text{MATLAB code based} \end{array} \right\} \text{Analytical + Numerical Solution}$$

$$\left. \begin{array}{l} \phi(x=0) = 0 \\ \phi(x=L) = 0 \end{array} \right\} \text{Boundary Conditions}$$

Bugs $\rightarrow$ -0.25 / simple bug

Send solution via email TGIERYA@ETHZ.CH

Analytical Solution:

integrate
twice eq.

$$\frac{\partial^2 \phi}{\partial x^2} = 2x^2 - \frac{x}{2} + e^x$$

$$\Rightarrow \int \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial x} \right) dx = \int 2x^2 dx - \int \frac{x}{2} dx + \int e^x dx$$

$$\Rightarrow \int \frac{\partial \phi}{\partial x} dx = \int 2 \cdot \frac{x^3}{3} dx - \frac{1}{2} \int \frac{x^2}{2} dx + \int e^x dx + \int C_1 dx$$

$$\Rightarrow \phi = \frac{2}{3} \cdot \frac{x^4}{4} - \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{x^3}{3} + e^x + C_1 x + C_2$$

$$\Rightarrow \boxed{\phi = \frac{1}{6} x^4 - \frac{1}{12} x^3 + e^x + C_1 x + C_2} \quad \text{General solution}$$

Boundary conditions:

$$\text{at } x=0 \Rightarrow \phi(x=0) = 0 = 0 + 0 + 1 + C_1 \cdot 0 + C_2 \Rightarrow C_2 = -1$$

$$\text{at } x=L \Rightarrow \phi(x=L) = 0 = \frac{L^4}{6} - \frac{L^3}{12} + e^L + C_1 L - 1$$

$$\Rightarrow C_1 = -L^3/6 + L^2/12 - e^L/L + 1/L$$

$$\boxed{\phi = \frac{1}{6} x^4 - \frac{1}{12} x^3 + e^x + x \left[-\frac{L^3}{6} + \frac{L^2}{12} + \frac{1-e^L}{L} \right] - 1}$$

Finite Difference solution

$$\frac{\partial^2 \phi}{\partial x^2} = 2x^2 - \frac{x}{2} + e^x$$

$$\text{LHS: } \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial x} \right)$$

$$= \frac{\phi_{i+1} + \phi_{i-1} - 2\phi_i}{\Delta x^2}$$

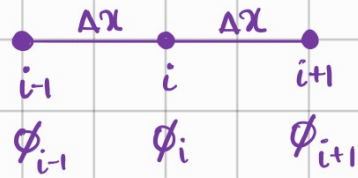
$$\begin{aligned} \text{RHS: } 2x^2 - 0.5x + e^x \\ = 2x_i^2 - 0.5x_i + e^{x_i} \end{aligned}$$

L Matrix:

$$L(i, i-1) = 1/\Delta x^2$$

$$L(i, i) = -2/\Delta x^2$$

$$L(i, i+1) = 1/\Delta x^2$$



R matrix:

$$R(i, 1) = 2x_i^2 - 0.5x_i + e^{x_i}$$

Boundary conditions:

Left $L(1, 1) = 1$ & $R(1, 1) = 0$
 $L(1, 2:N) = 0$

Right $L(N, N) = 1$ & $R(N, 1) = 0$
 $L(N, 1:N-1) = 0$